

Intelligent Fabric Defect Detection Using Machine Learning and Deep Learning Techniques

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Abstract—Fabric defect detection is an important task in the textile industry for maintaining product quality and reducing manufacturing losses. Manual inspection methods are often time-consuming, labor-intensive, and prone to human error. This project presents an intelligent fabric defect detection system based on machine learning and deep learning techniques. A multi-class fabric defect dataset containing nine fabric conditions was utilized for model development and evaluation. Histogram of Oriented Gradients (HOG), Color Histogram Features, and Spatial Binning Features were extracted and combined to improve classification performance. Multiple machine learning algorithms including Naive Bayes, Support Vector Machine (SVM), Logistic Regression, and Multi-Layer Perceptron (MLP) were trained and evaluated. In addition, a Convolutional Neural Network (CNN) was implemented for performance comparison. Experimental results demonstrated that the MLP model trained on combined features achieved the highest accuracy of 90.55

Keywords: Fabric Defect Detection, Machine Learning, Deep Learning, Computer Vision, Image Classification, HOG Features, Color Histogram, Spatial Binning, Multi-Layer Perceptron, CNN

I. INTRODUCTION

FABRIC Quality assurance is a critical requirement in the textile industry, where fabric defects can significantly affect product quality, customer satisfaction, and manufacturing costs. Common fabric defects such as holes, stains, broken stitches, needle marks, horizontal defects, and vertical defects reduce the commercial value of textile products and may lead to financial losses for manufacturers. Traditionally, fabric inspection is performed manually by human operators; however, manual inspection is often slow, labor-intensive, and susceptible to human error.

Recent advancements in Machine Learning (ML), Computer Vision, and Deep Learning (DL) have enabled the development of intelligent automated inspection systems capable of detecting and classifying fabric defects with high accuracy. These systems analyze fabric images, extract meaningful visual features, and automatically identify defect categories without requiring continuous human supervision. Automated defect detection not only improves inspection speed but also increases consistency and reliability in quality control processes.

In this project, an intelligent fabric defect detection system was developed using both machine learning and deep learning approaches. The proposed methodology utilizes Histogram of Oriented Gradients (HOG), Color Histogram Features, and Spatial Binning Features to capture texture, color distribution, and spatial information from fabric images. Multiple

machine learning models including Naive Bayes, Support Vector Machine (SVM), Logistic Regression, and Multi-Layer Perceptron (MLP) were trained and evaluated. Furthermore, a Convolutional Neural Network (CNN) was implemented to provide a deep learning-based comparison.

To demonstrate practical applicability, a graphical user interface (GUI) was developed using Python Tkinter. The developed application enables users to upload fabric images, perform automatic defect classification, generate PDF reports, and visualize model performance statistics. The primary objective of this research is to investigate the effectiveness of machine learning and deep learning techniques for automated fabric defect detection and identify the most suitable approach for real-world textile inspection systems.

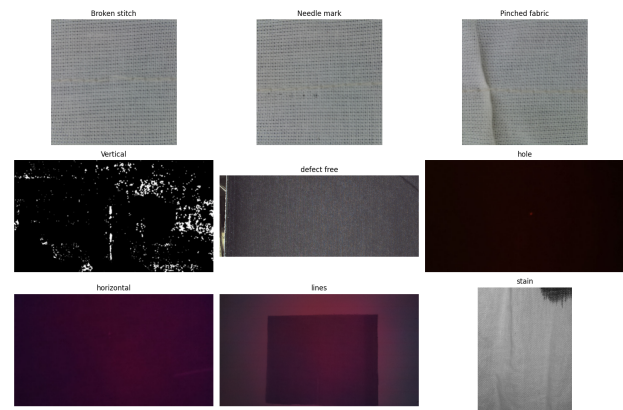


Fig. 1. Sample Images from the Fabric Defect Dataset

II. LITERATURE REVIEW

Fabric defect detection has been an active research area in the fields of computer vision, image processing, and machine learning. Traditional fabric inspection methods primarily relied on manual observation by human inspectors. Although manual inspection can identify visible defects, it is often slow, costly, and highly dependent on operator experience. Furthermore, human inspection becomes less reliable when large volumes of fabric must be examined continuously.

To overcome these limitations, researchers introduced image processing techniques for automated defect detection. Early approaches focused on edge detection, thresholding, texture analysis, and statistical image processing methods. These techniques provided basic automation but often struggled when

dealing with complex fabric textures and varying illumination conditions.

With the advancement of machine learning, various classification algorithms such as Naive Bayes, Support Vector Machine (SVM), Decision Trees, Random Forests, and Artificial Neural Networks were applied to fabric defect detection problems. These approaches typically rely on handcrafted features extracted from images to represent texture, color, and structural information. Feature extraction methods such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and color-based descriptors have shown promising results in textile inspection applications.

More recently, deep learning techniques, particularly Convolutional Neural Networks (CNNs), have gained significant attention due to their ability to automatically learn hierarchical image features. CNN-based approaches have demonstrated strong performance in various image classification and object detection tasks, including fabric defect detection. However, deep learning models often require large computational resources, extensive training data, and longer training times.

Despite the success of deep learning approaches, machine learning methods combined with effective feature engineering remain attractive for practical industrial applications due to their lower computational cost and easier deployment. Therefore, this project investigates both machine learning and deep learning approaches and compares their performance for automated fabric defect detection using a multi-class fabric defect dataset.

TABLE I
SUMMARY OF RELATED STUDIES

Study	Method	Outcome
Study 1	Traditional Image Processing	Moderate Accuracy
Study 2	Machine Learning	Improved Performance
Study 3	Deep Learning (CNN)	High Accuracy

III. DATASET DESCRIPTION

The fabric defect dataset used in this project consists of nine different fabric conditions, including both defective and defect-free samples. The dataset was collected and organized into separate class folders to facilitate supervised machine learning training and evaluation.

The dataset contains a total of 3,067 images distributed across nine classes. These classes include Broken Stitch, Needle Mark, Pinched Fabric, Vertical Defect, Defect Free, Hole, Horizontal Defect, Lines, and Stain. The images are available in JPG and PNG formats and represent various fabric textures and defect patterns.

Before model training, all images were resized to a fixed resolution and processed to ensure consistency during feature extraction. The dataset was later divided into training and testing subsets for model evaluation and comparison.

Table II summarizes the overall dataset statistics, while Figure 2 illustrates the distribution of images across different fabric defect categories.

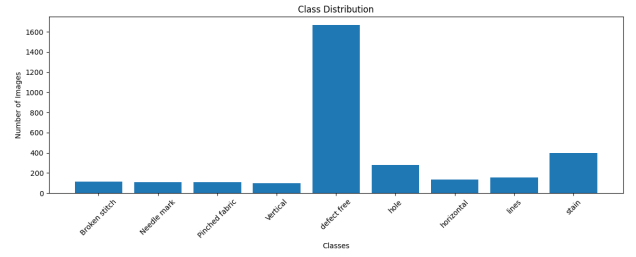


Fig. 2. Class Distribution of the Fabric Defect Dataset

TABLE II
DATASET STATISTICS

Attribute	Value
Total Images	3067
Number of Classes	9
Training Images	2453
Testing Images	614
Corrupted Images	0
Image Formats	JPG, PNG

IV. METHODOLOGY

The proposed fabric defect detection system was developed using a structured machine learning and deep learning workflow. The objective of the methodology was to accurately identify fabric defects by extracting meaningful visual features from fabric images and evaluating multiple classification algorithms. The complete process consisted of dataset acquisition, preprocessing, feature extraction, feature fusion, model training, model evaluation, and graphical user interface deployment.

Initially, the dataset was collected and analyzed to understand class distributions and image characteristics. Various preprocessing operations were then applied to improve data quality and ensure consistency across all samples. Multiple feature extraction techniques were investigated to capture texture, color, and spatial information from fabric images. These extracted features were subsequently used to train several machine learning classifiers. The best-performing classifier was further evaluated using different feature extraction methods. Finally, a deep learning-based Convolutional Neural Network (CNN) was implemented and compared against the best machine learning model.

Figure 3 illustrates the overall workflow adopted in this study.

A. Data Collection

The dataset utilized in this project consists of fabric images representing both defect-free and defective fabric samples. The images were organized into separate folders corresponding to individual class labels. A total of nine classes were included in the dataset, namely Broken Stitch, Needle Mark, Pinched Fabric, Vertical Defect, Defect Free, Hole, Horizontal Defect, Lines, and Stain.

The collected dataset contained 3,067 images with varying defect characteristics and texture patterns. Since the objective

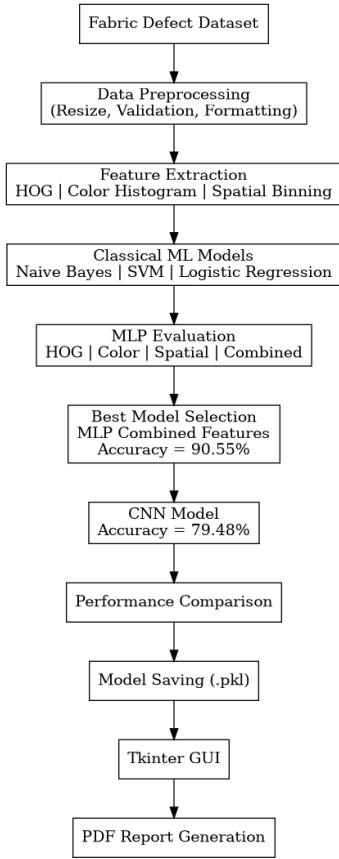


Fig. 3. Overall Workflow of the Proposed Fabric Defect Detection System

of the study was multi-class classification, the dataset was carefully organized to ensure that each image was assigned the correct class label. The availability of both defective and non-defective samples allowed the system to learn meaningful distinctions between normal and abnormal fabric conditions.

B. Data Preprocessing

Data preprocessing was performed before feature extraction and model training. The primary purpose of preprocessing was to improve image consistency and remove potential issues that could negatively impact model performance.

Initially, all images were validated to ensure that they could be successfully loaded into memory. Corrupted or unreadable files were identified and removed. The dataset was then analyzed to determine image dimensions, image formats, and class distributions.

To maintain consistency throughout the feature extraction process, all images were resized to a fixed resolution of 224x224 pixels. Standardizing image dimensions ensured that each image generated feature vectors of equal length. In addition, grayscale conversion was applied whenever required by specific feature extraction techniques such as Histogram of Oriented Gradients (HOG).

These preprocessing operations reduced variability within the dataset and provided a standardized input representation for subsequent processing stages.

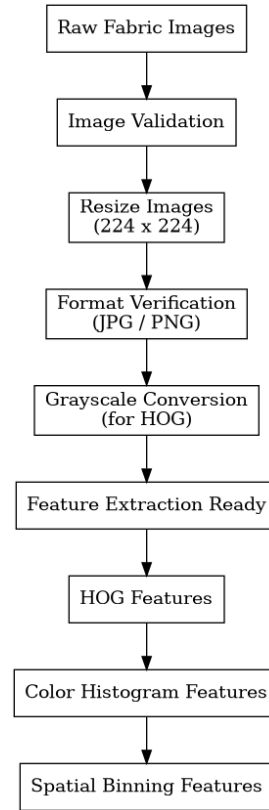


Fig. 4. Image Preprocessing Pipeline

C. Feature Extraction

Feature extraction is one of the most critical stages of any machine learning-based image classification system. Instead of directly providing raw images to machine learning models, meaningful numerical representations were extracted from each image. These representations captured texture information, color characteristics, and spatial patterns present within fabric samples.

Three different feature extraction techniques were investigated in this study:

- Histogram of Oriented Gradients (HOG)
- Color Histogram Features
- Spatial Binning Features

The purpose of employing multiple feature extraction methods was to evaluate their individual effectiveness and later combine them into a unified feature representation.

1) *Histogram of Oriented Gradients (HOG)*: Histogram of Oriented Gradients (HOG) is a widely used feature extraction technique in computer vision applications. HOG focuses on the distribution of edge directions and gradient orientations within an image. Since many fabric defects alter local texture structures, HOG is particularly effective in capturing defect-related patterns.

For HOG extraction, each image was first converted to grayscale. Gradient information was then computed, and orientation histograms were generated over localized image re-

gions. The resulting feature vector contained 26,244 numerical features per image.

The extracted HOG features provided a detailed representation of fabric textures and were later used for machine learning classification.

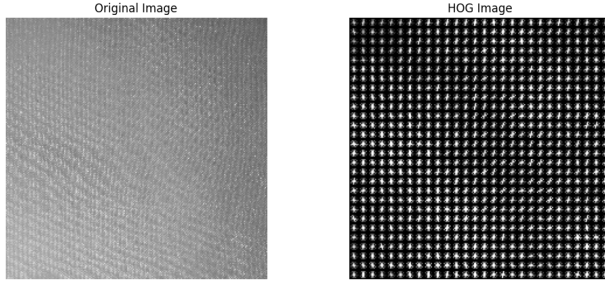


Fig. 5. Visualization of HOG Features

2) *Color Histogram Features*: Color Histogram features were extracted to represent the color distribution present within fabric images. Unlike HOG features, which primarily capture texture information, color histograms describe variations in pixel intensities across RGB color channels.

Histograms were computed using 32 bins per color channel. The resulting histograms were normalized and transformed into a one-dimensional feature vector. This representation proved useful for identifying defects associated with color abnormalities such as stains and discoloration.

The generated Color Histogram feature vector contained 32,768 features per image.

TABLE III
COLOR HISTOGRAM FEATURE STATISTICS

Metric	Value
Feature Length	32768
Training Shape	(2453,32768)
Testing Shape	(614,32768)
Accuracy	84.04

3) *Spatial Binning Features*: Spatial Binning is a simple yet effective feature extraction technique that preserves coarse spatial information from images. In this approach, images were resized to a lower resolution and then flattened into a one-dimensional vector.

Although Spatial Binning does not capture detailed texture information, it retains overall structural and spatial characteristics of fabric images. The extracted feature vector contained 3,072 features per image.

Experimental results demonstrated that Spatial Binning provided competitive performance despite its simplicity.

D. Feature Combination

After evaluating individual feature extraction techniques, a feature fusion strategy was adopted. The motivation behind feature combination was to exploit the strengths of all feature extraction methods simultaneously.

TABLE IV
SPATIAL BINNING FEATURE STATISTICS

Metric	Value
Feature Length	3072
Training Shape	(2453,3072)
Testing Shape	(614,3072)
Accuracy	85.50

HOG features contributed texture information, Color Histogram features provided color distribution information, and Spatial Binning preserved global spatial characteristics. By combining all three representations, a richer and more discriminative feature vector was obtained.

The final combined feature vector contained 62,084 features per image and served as the input for subsequent machine learning experiments.

TABLE V
COMBINED FEATURE STATISTICS

Feature Type	Length
HOG Features	26244
Color Histogram Features	32768
Spatial Binning Features	3072
Total Combined Features	62084

E. Train-Test Split

To evaluate model performance objectively, the dataset was divided into training and testing subsets using an 80:20 ratio. The training subset was used for learning model parameters, whereas the testing subset was reserved exclusively for performance evaluation.

Out of the total 3,067 images, 2,453 images were allocated for training and 614 images were allocated for testing. This split ensured that models were evaluated on previously unseen data and provided a realistic estimate of generalization performance.

TABLE VI
TRAINING AND TESTING DATASET DISTRIBUTION

Dataset	Number of Images
Training Set	2453
Testing Set	614
Total	3067

V. MACHINE LEARNING MODELS

After completing data preprocessing and feature extraction, multiple machine learning algorithms were trained and evaluated for fabric defect classification. The primary objective was to determine which classification algorithm could most effectively distinguish between different fabric defect categories. Since fabric images contain complex texture patterns and visual variations, evaluating multiple classifiers was necessary to identify the most suitable model.

Initially, three traditional machine learning algorithms were trained using the extracted feature vectors. These algorithms

included Naive Bayes, Support Vector Machine (SVM), and Logistic Regression. Their performance was compared using classification accuracy and confusion matrices. After identifying the most promising classification strategy, Multi-Layer Perceptron (MLP) was further evaluated using different feature extraction methods, including HOG, Color Histogram, Spatial Binning, and Combined Features.

The following subsections discuss each classifier in detail.

A. Naive Bayes

Naive Bayes is a probabilistic machine learning algorithm based on Bayes' Theorem. The classifier assumes that all input features are statistically independent of one another. Although this assumption is rarely true in real-world image datasets, Naive Bayes remains computationally efficient and is often used as a baseline classification model.

In this project, Naive Bayes was trained using the extracted image features and evaluated on the testing dataset. Due to the complexity of fabric textures and the strong correlation between image features, the independence assumption limited the model's ability to accurately distinguish between defect categories.

Experimental results showed that Naive Bayes achieved an overall classification accuracy of 56.84. Although the model successfully identified some defect classes, its performance was significantly lower than the other evaluated classifiers. The obtained results indicate that probabilistic models alone may not be sufficient for handling complex texture-based classification problems.

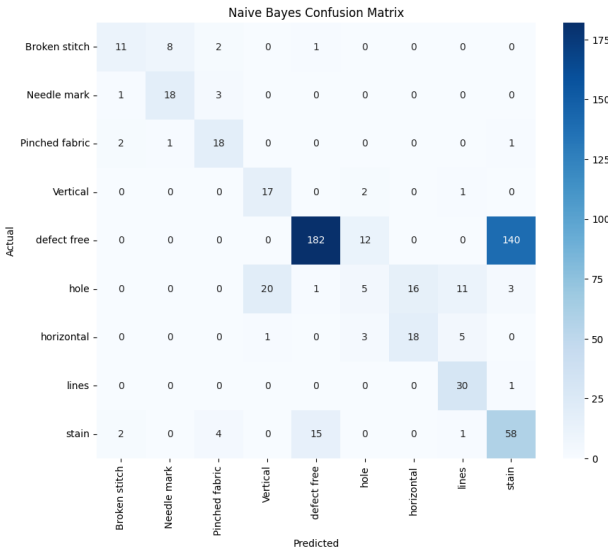


Fig. 6. Naive Bayes Confusion Matrix

Figure 6 presents the confusion matrix of the Naive Bayes classifier. The matrix indicates that the model experienced considerable difficulty in correctly separating several defect categories. A significant number of misclassifications can be observed across multiple classes, which contributed to the relatively low classification accuracy of 56.84. The results

suggest that the probabilistic assumptions of Naive Bayes were insufficient for capturing the complex relationships present within fabric defect features.

B. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm that constructs optimal decision boundaries between classes. The primary objective of SVM is to maximize the margin between different class boundaries, resulting in improved generalization performance.

SVM is particularly effective for high-dimensional feature spaces and has been widely used in image classification applications. Since the extracted fabric features contained a large number of dimensions, SVM was considered a suitable candidate for classification.

After training and evaluation, the SVM classifier achieved an accuracy of 85.34. This represented a significant improvement over Naive Bayes. The model successfully learned discriminative boundaries between defect classes and demonstrated strong classification performance across multiple categories.

The obtained results indicate that SVM is highly effective for fabric defect classification when combined with appropriate feature extraction techniques.

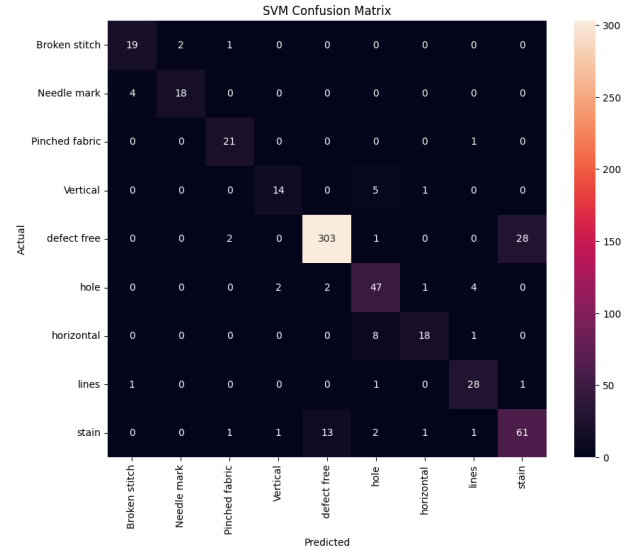


Fig. 7. Support Vector Machine Confusion Matrix

Figure 7 illustrates the confusion matrix obtained using the Support Vector Machine classifier. Compared with Naive Bayes, SVM achieved substantially better class separation and reduced the number of classification errors. Most defect categories were correctly identified, demonstrating the effectiveness of SVM in handling high-dimensional feature spaces. The improved confusion matrix is consistent with the higher classification accuracy of 85.34.

C. Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability of an image belonging to a particular

class. Despite its simplicity, Logistic Regression remains a powerful machine learning technique for many classification tasks.

In this study, Logistic Regression was trained using the extracted fabric features and evaluated on unseen testing data. The model demonstrated stable classification performance and successfully captured relationships between extracted image features and defect labels.

Experimental evaluation showed that Logistic Regression achieved an accuracy of 85.83% slightly outperforming the SVM classifier. The results indicate that even relatively simple classification models can achieve competitive performance when informative image features are available.



Fig. 8. Logistic Regression Confusion Matrix

Figure 8 presents the confusion matrix of the Logistic Regression classifier. The model demonstrated strong classification performance across multiple defect categories and achieved an overall accuracy of 85.83%. The confusion matrix reveals that most samples were correctly classified, although a limited number of misclassifications remained for visually similar defect classes. Overall, Logistic Regression provided competitive performance despite its relatively simple linear decision boundaries.

TABLE VII
PERFORMANCE OF CLASSICAL MACHINE LEARNING MODELS

Model	Accuracy (%)
Naive Bayes	56.84
Support Vector Machine (SVM)	85.34
Logistic Regression	85.83

The comparison presented in Table VII shows that both SVM and Logistic Regression substantially outperformed Naive Bayes. Consequently, further experimentation focused on improving classification performance through neural network-based approaches.

D. Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) is a feed-forward artificial neural network capable of learning complex non-linear relationships between input features and output classes. Unlike traditional machine learning algorithms, MLP contains multiple interconnected neurons organized into input, hidden, and output layers.

The motivation for using MLP in this project was its ability to learn complex feature interactions that may not be captured by simpler classifiers. Since fabric defect images contain intricate texture patterns, color variations, and structural irregularities, neural network-based models provide greater flexibility for classification.

Rather than evaluating MLP using only a single feature extraction technique, multiple experiments were conducted using different feature representations. The objective was to determine which feature extraction strategy provided the most informative representation of fabric defects.

The first experiment utilized HOG features as input to the MLP classifier. Since HOG captures texture information and edge orientations, strong classification performance was expected. The resulting model achieved an accuracy of 88.27.

The second experiment employed Color Histogram features. These features describe the distribution of pixel intensities across RGB channels and are particularly useful for identifying color-related defects such as stains. The MLP classifier achieved an accuracy of 84.04.

The third experiment utilized Spatial Binning features. These features preserve coarse spatial information and global image structure. The resulting model achieved an accuracy of 85.50.

Finally, all three feature extraction methods were combined into a single hybrid feature representation. The combined feature vector integrated texture information from HOG, color information from Color Histograms, and structural information from Spatial Binning. This comprehensive representation significantly improved classification performance.

The MLP classifier trained using Combined Features achieved the highest overall accuracy of 90.55%, making it the best-performing machine learning model in this study.

Figure 9 shows the confusion matrix of the Multi-Layer Perceptron classifier. The matrix demonstrates significantly improved classification performance compared with the previously evaluated machine learning models. Most defect categories were correctly recognized with relatively few classification errors. The improved distribution of correctly predicted samples highlights the ability of MLP to learn complex non-linear relationships within fabric defect features. This strong confusion matrix performance contributed to the highest overall accuracy achieved by the proposed machine learning framework.

The results presented in Table VIII clearly demonstrate the effectiveness of feature fusion. The combined feature representation outperformed all individual feature extraction methods and provided the most discriminative description of fabric defects.

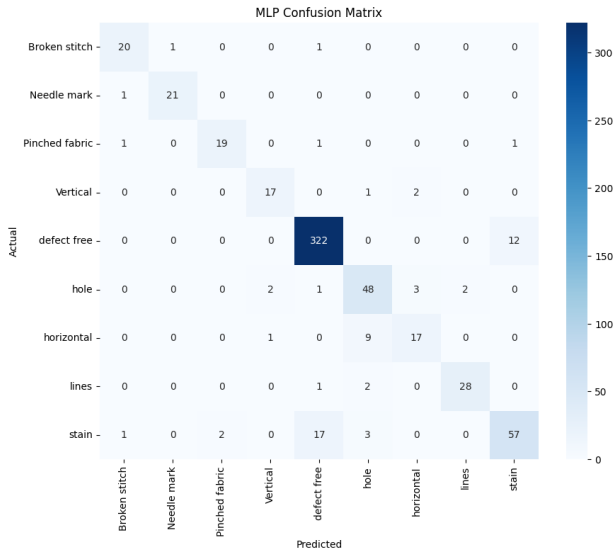


Fig. 9. MLP Confusion Matrix

TABLE VIII
MLP PERFORMANCE USING DIFFERENT FEATURE EXTRACTION METHODS

Feature Extraction Method	Accuracy (
HOG Features	88.27
Color Histogram Features	84.04
Spatial Binning Features	85.50
Combined Features	90.55

VI. CNN-BASED COMPARISON

Although traditional machine learning models demonstrated promising results, recent advancements in Artificial Intelligence and Computer Vision have shown that Deep Learning models can automatically learn complex image representations without requiring manual feature engineering. Therefore, a Convolutional Neural Network (CNN) was implemented in this project to compare the performance of deep learning with the previously developed machine learning approach.

CNNs have become one of the most widely used architectures for image classification tasks because they are capable of automatically extracting hierarchical features directly from raw image data. Unlike traditional machine learning models that depend on manually engineered features such as HOG, Color Histograms, and Spatial Binning, CNNs learn feature representations automatically during training.

The primary objective of this experiment was to determine whether automatic feature learning could outperform handcrafted feature engineering for fabric defect classification.

A. CNN Architecture

The Convolutional Neural Network developed in this study consisted of multiple layers designed to progressively learn visual patterns from fabric images.

The architecture included:

- Convolutional Layers

- ReLU Activation Functions
- Max Pooling Layers
- Flatten Layer
- Fully Connected Dense Layers
- Softmax Output Layer

The convolutional layers were responsible for detecting local image features such as edges, textures, defect boundaries, and structural irregularities. Pooling layers reduced spatial dimensions and computational complexity while preserving important information. The fully connected layers performed the final classification by mapping extracted features to one of the nine fabric classes.

The output layer contained nine neurons corresponding to the nine fabric defect categories included in the dataset.

B. CNN Training Process

Before CNN training, all images were resized to a fixed resolution of 224×224 pixels. The images were then normalized to ensure numerical stability during optimization.

The dataset was divided into training and testing subsets using the same 80:20 ratio employed throughout the machine learning experiments. This ensured fair comparison between CNN and traditional machine learning models.

During training, the CNN automatically learned feature representations from fabric images. Unlike HOG, Color Histogram, and Spatial Binning methods, no manual feature extraction was required. The network iteratively adjusted its weights using backpropagation and gradient descent to minimize classification error.

The training process enabled the network to learn visual patterns associated with different defect categories, including holes, stains, broken stitches, needle marks, horizontal defects, vertical defects, and other fabric abnormalities.

C. CNN Performance Evaluation

After training was completed, the CNN model was evaluated on the testing dataset. Classification accuracy was used as the primary evaluation metric.

Experimental results showed that the CNN achieved an overall classification accuracy of 79.48

TABLE IX
CNN PERFORMANCE RESULTS

Metric	Value
Model	CNN
Accuracy	79.48
Number of Classes	9
Image Resolution	224×224

Although the CNN successfully learned meaningful visual patterns from the fabric images, its performance remained lower than the best-performing machine learning approach based on handcrafted features.

D. Comparison with Combined Features Model

The final stage of experimentation involved comparing the CNN model against the best machine learning model developed in this study.

Among all machine learning experiments, the MLP classifier trained using Combined Features achieved the highest classification accuracy of 90.55%. The Combined Features approach integrated information from HOG descriptors, Color Histogram features, and Spatial Binning features into a single hybrid feature vector containing 62,084 features.

The comparison between the two approaches is presented in Table X.

TABLE X
COMPARISON BETWEEN COMBINED FEATURES AND CNN

Model	Accuracy (%)
MLP (Combined Features)	90.55
CNN	79.48

The results clearly demonstrate that the machine learning approach based on combined handcrafted features outperformed the CNN model by a considerable margin. The MLP classifier achieved approximately 11% higher accuracy than the CNN model.

Several factors may have contributed to this outcome. First, the dataset size was relatively limited for deep learning applications. CNNs typically require significantly larger datasets to achieve optimal performance. Second, handcrafted features such as HOG, Color Histograms, and Spatial Binning were specifically designed to capture important fabric characteristics, thereby providing highly informative representations for classification.

The findings of this study suggest that carefully engineered feature extraction methods can remain highly competitive, particularly when training data is limited. Furthermore, machine learning models generally require fewer computational resources and shorter training times compared to deep learning approaches.

Therefore, the MLP classifier trained on Combined Features was selected as the final model for deployment within the proposed Fabric Defect Detection System.

VII. GUI DEVELOPMENT (TKINTER)

To provide a user-friendly environment for practical deployment, a graphical user interface (GUI) was developed using Python Tkinter. The application allows users to upload fabric images and automatically classify defects using the trained machine learning model.

The GUI integrates the saved model, feature extraction pipeline, label encoder, and scaler to generate real-time predictions. Users can easily select images, view classification results, and interact with the system without requiring technical knowledge of machine learning.

The main functionalities of the GUI include:

- Upload Fabric Image
- Automatic Defect Classification

- Fabric Status Identification
- Model Statistics Visualization
- PDF Report Generation

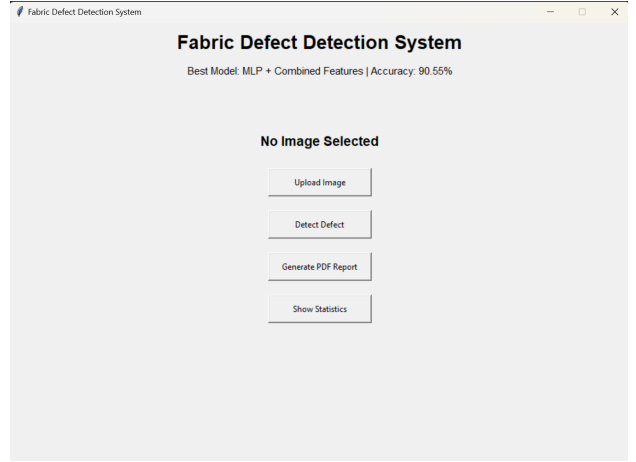


Fig. 10. Main Graphical User Interface

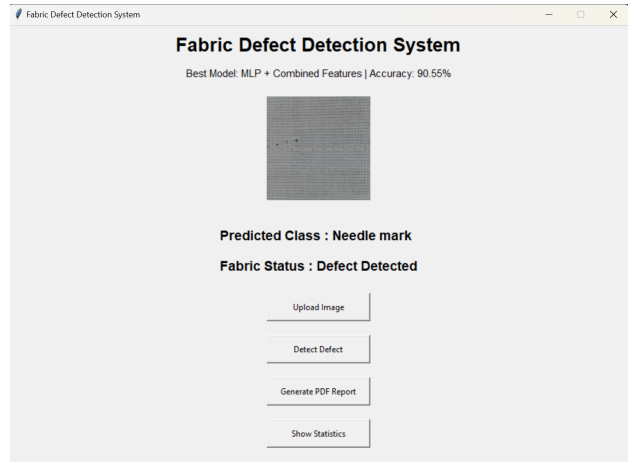


Fig. 11. Defect Prediction Result

VIII. PDF REPORT GENERATION

An automated PDF reporting module was integrated into the application using the ReportLab library. After performing defect classification, users can generate a PDF report containing prediction results, fabric status information, model details, and system performance metrics.

The reporting functionality improves documentation and provides a convenient method for storing inspection results. This feature enhances the practical applicability of the developed system in industrial environments.

IX. CONCLUSION

This project presented an intelligent fabric defect detection system using machine learning and deep learning techniques. Multiple feature extraction methods including Histogram of Oriented Gradients (HOG), Color Histogram Features, and

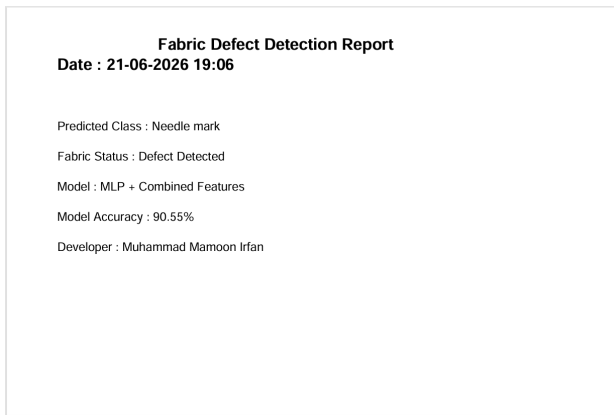


Fig. 12. Generated PDF Inspection Report

Spatial Binning Features were utilized to represent fabric images effectively. Several machine learning models and a Convolutional Neural Network (CNN) were trained and evaluated.

Experimental results demonstrated that the Multi-Layer Perceptron (MLP) trained on combined features achieved the highest classification accuracy of 90.55

X. FUTURE WORK

Several improvements can be incorporated in future versions of the proposed system. Advanced deep learning architectures such as YOLO, EfficientDet, ResNet, and Vision Transformers can be explored for improved defect localization and classification. Real-time defect detection using live camera feeds may also be implemented for industrial deployment.

Future work may additionally include defect localization through bounding boxes, cloud-based monitoring systems, mobile application integration, and support for larger industrial textile datasets to further improve robustness and scalability.

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PROJECT RESOURCES

Overleaf Documentation:

<https://www.overleaf.com/read/jzdgxgkcv#44372b>

Google Colab Implementation:

https://colab.research.google.com/drive/1fFK3lZwDCiK_bDsTtlcCnv6f3jkgu3Nd?usp=sharing

Project Presentation Slides:

https://drive.google.com/drive/folders/1iCyreoCPyeXy_2Sc3thyb91xox2Fpo?usp=sharing

Front-End Application and Supporting Files:

<https://drive.google.com/drive/folders/1IVhz-G565Tx2KdapyBmbPS-T7YP-NZb3?usp=sharing>

Project Demonstration Video:

https://www.youtube.com/live/iu1aycB7EUK?si=WqBTeh3M0pivsY_a



MUHAMMAD MAMOON IRFAN (Group Leader) is a BS Computer Science student at the University of Management and Technology (UMT), Lahore. His research interests include Machine Learning, Artificial Intelligence, Computer Vision, Data Science, and Software Development. He served as the team leader for this project and was responsible for model development, feature engineering, system integration, and performance evaluation.



ASFAND AMJAD is a BS Computer Science student at the University of Management and Technology (UMT), Lahore. His interests include AI Engineering Machine Learning, Software Engineering, computer vision and Data Analysis. He contributed to dataset preparation, experimentation, and project documentation.